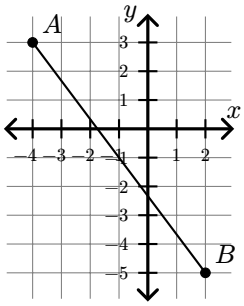
Section 6.1 — Coordinate Geometry — Answer Key

Math 103: College Algebra

Practice With Points

1. On a separate sheet of graph paper (or the other side of this sheet), draw appropriate axes and:

a.



b. There are two possible diagrams you could use here.

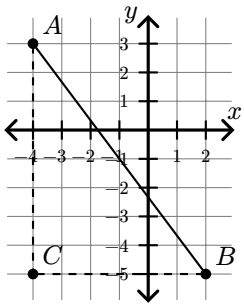


Diagram I

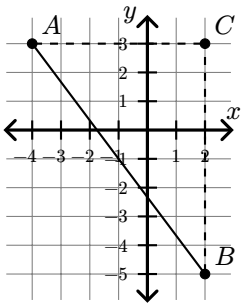


Diagram II

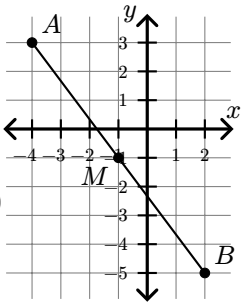
c. With Diagram I: $C = (-4, -5)$
With Diagram II: $C = (2, 3)$

d. With Diagram I: $AC = 8$
With Diagram II: $AC = 6$

e. With Diagram I: $BC = 6$
With Diagram II: $BC = 8$

f. $AB = 10$

g. $M = (-1, 1)$



Distance Formula

Using the distance formula (without plotting on a graph) find the distance between the two points given. If necessary, give your answer as a simplified radical and as a decimal approximation.

2. 13
3. 5
4. $\sqrt{5} \approx 2.236$
5. 2.9
6. 1
7. $\frac{2\sqrt{298}}{9} \approx 3.836$

Midpoint Formula

For each pair of points A and B given below, find the midpoint of $\overline{AB}$.

8. $M = (2, 2)$
9. $M = (8, 4.5)$
10. $M = (-8.5, -4)$
11. $M = \left(\frac{5}{2}, 3\right)$
12. $M = (1.3, 3.7)$
13. $M = (-38.5, 60.5)$

Triangles and Inequalities

For each list of three values, classify them as one of the following.

A. Not possible as the distances between three points.

B. Distances between three collinear points.

C. Side lengths of an obtuse triangle.

D. Side lengths of a right triangle.

E. Side lengths of an acute triangle.

14. C	15. B	16. E	17. A	18. D
19. B	20. E	21. C	22. A	23. D

Classifying Triangles

Use the distance formula to classify each set of three points as:

A. Three collinear points

B. Corners of an obtuse triangle

C. Corners of a right triangle

D. Corners of an acute triangle

24. A	25. C	26. B	27. B